



A Probabilistic Ranking Model for Large-Scale Athletic Performance

Sophie Luxton¹, Abigail Mabe¹, Coleman Ferrell¹, Mario Giacomazzo¹

¹Department of Statistics and Operations Research, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599



Introduction

Ranking athletes is challenging because many competitors never race each other directly. Traditional ranking systems rely primarily on finishing times, which fail to account for differences in competition strength, race conditions, and tactical dynamics.

Goal

Develop a ranking system that reflects *who you beat*, not just how fast you run, by incorporating competitive interactions between athletes.

Key Idea

Athletic performance should be evaluated within the context of competition networks rather than isolated time-based outcomes.

Why Not Times?

Time – 4:10	Athlete A (weak race)
Reality: 4:12	Athlete B (strong race)
B beats stronger competitors → actually better	

Time-based rankings can misrepresent performance when competition strength differs.

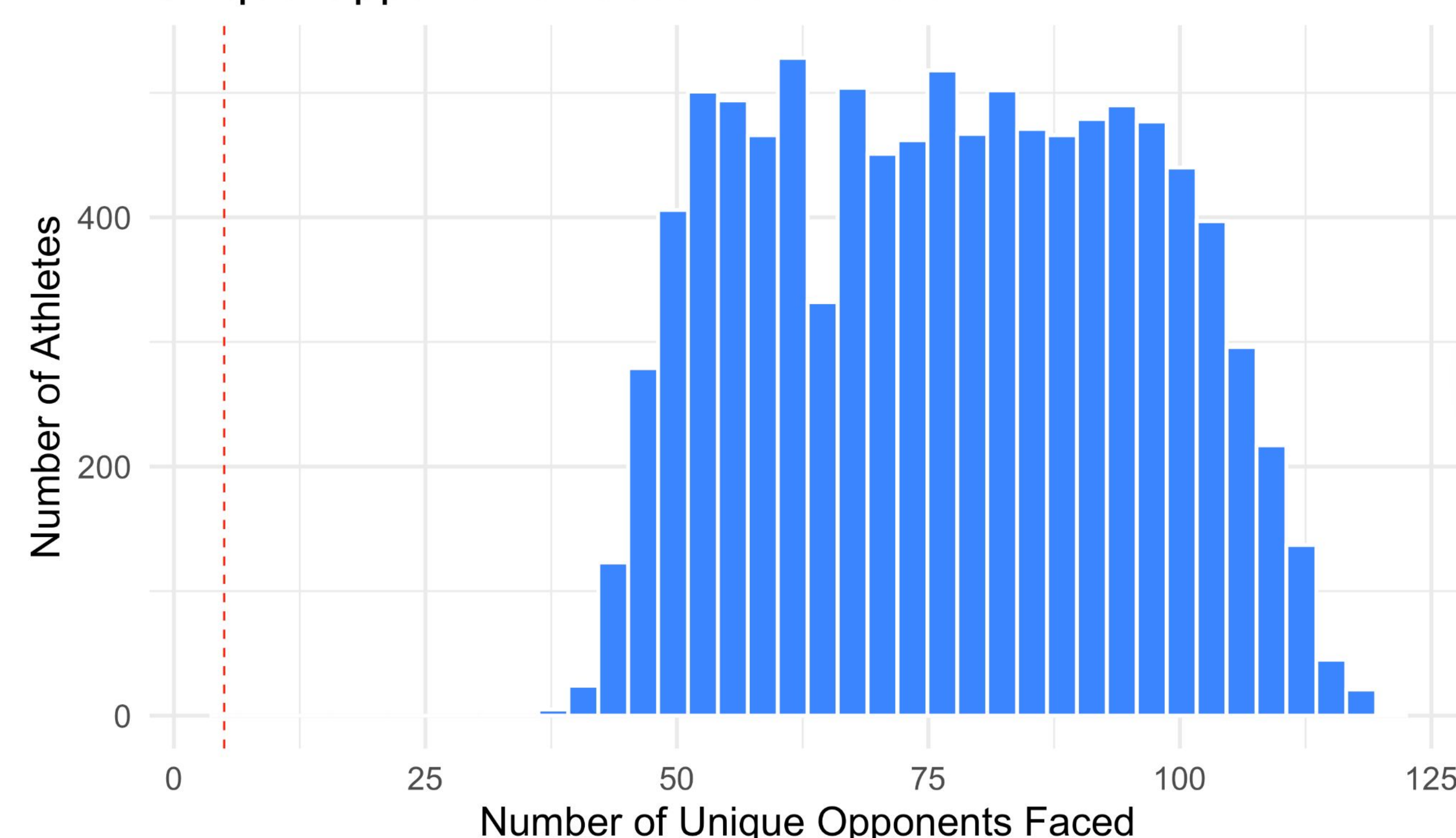
Modern day athletics is not just about someone's raw speed but also their ability to race. Focusing solely on raw time data is unfair when we are ranking large data sets as conditions are different at every meet and track.

Data

We simulated data for the women's 1500m with the following specifications:

- 10,000 individual athletes
- Each athlete racing five to ten times totaling 6,816 races
- 74,874 individual observations
- Each observation represents one athlete's time
- We have columns for race id, athlete id, finish position, time in seconds, event and sex
- Our finish time in seconds was created by using a truncated normal with a mean of 281 and a standard deviation of 16

Unique Opponents Faced Per Athlete



Methods

Luce's Axiom of Choice Luce's Axiom of Choice states that the probability of choosing an option depends only on how strong it is relative to the other available choices. Furthermore, this probability remains unchanged if irrelevant alternatives are removed.

Plackett–Luce Model The Plackett–Luce model estimates the latent ability of each athlete from observed race rankings.

Model Setup Suppose we have j unique athletes in a given race, represented as the set $S = \{i_1, i_2, \dots, i_j\}$

Each athlete $i \in S$ is assigned a latent ability parameter α_i

The probability that athlete j wins the race is:

$$P(j | S) = \frac{\alpha_j}{\sum_{i \in S} \alpha_i}$$

That is, their strength divided by the sum of all athletes' strengths in the field.

Now suppose a race produces the finishing order $i_1 \succ i_2 \succ \dots \succ i_j$ where i_1 is the winner, i_2 finishes second, and so on. The probability of observing this full ranking is:

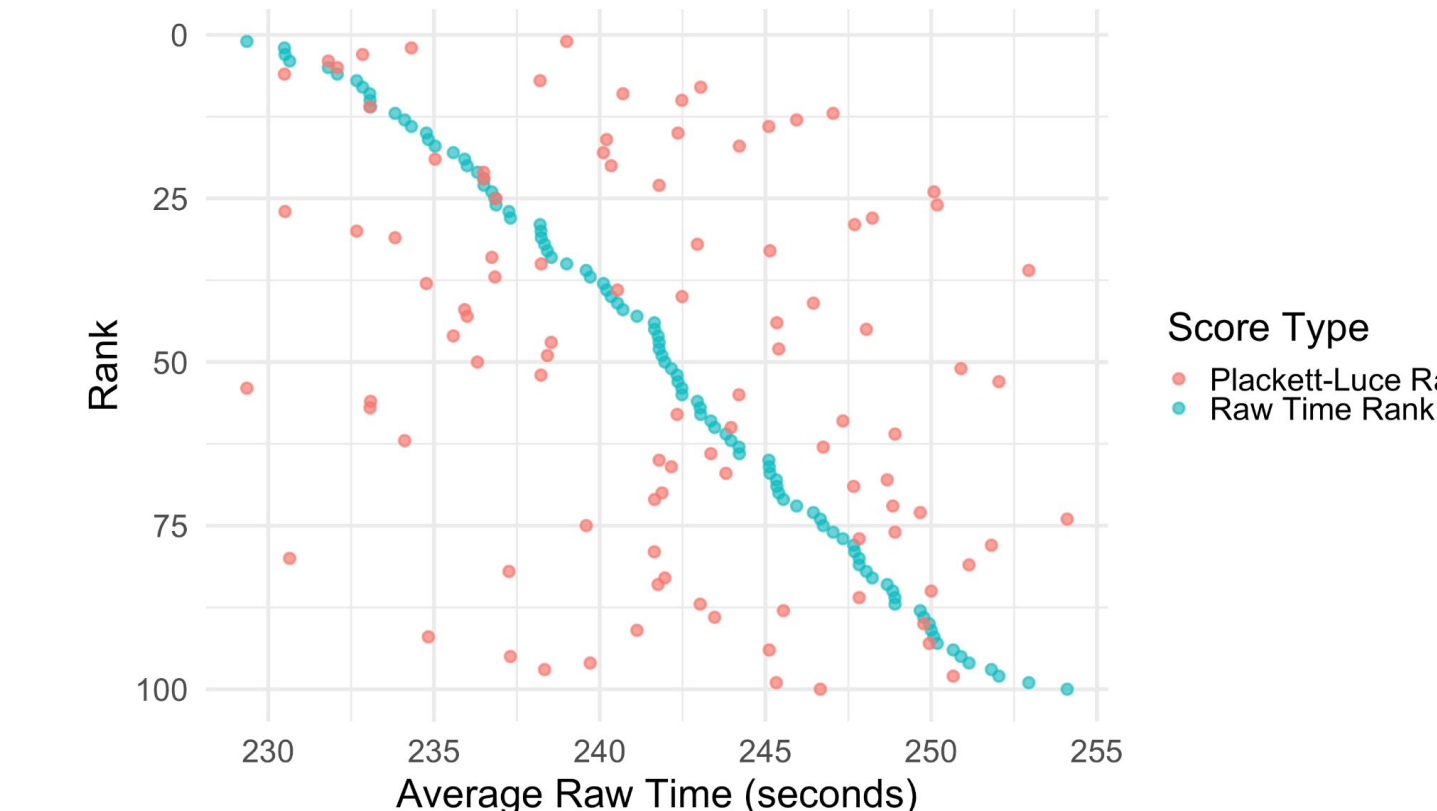
$$P(i_1 \succ i_2 \succ \dots \succ i_j) = \prod_j \frac{\alpha_{i_j}}{\sum_{i \in A_j} \alpha_i}$$

where $A_j = \{i_j, i_{j+1}, \dots, i_j\}$ is the set of athletes still remaining at stage r . At each stage, the denominator contains only athletes who have not yet finished.

Intuition The model finds the probability that one athlete wins given everyone in the race, multiplied by the probability the second-place athlete wins given all remaining athletes, and so on down the finishing order. By fitting these probabilities across thousands of races, the model returns a single strength score α_i per athlete — enabling fair comparison across races with different fields.

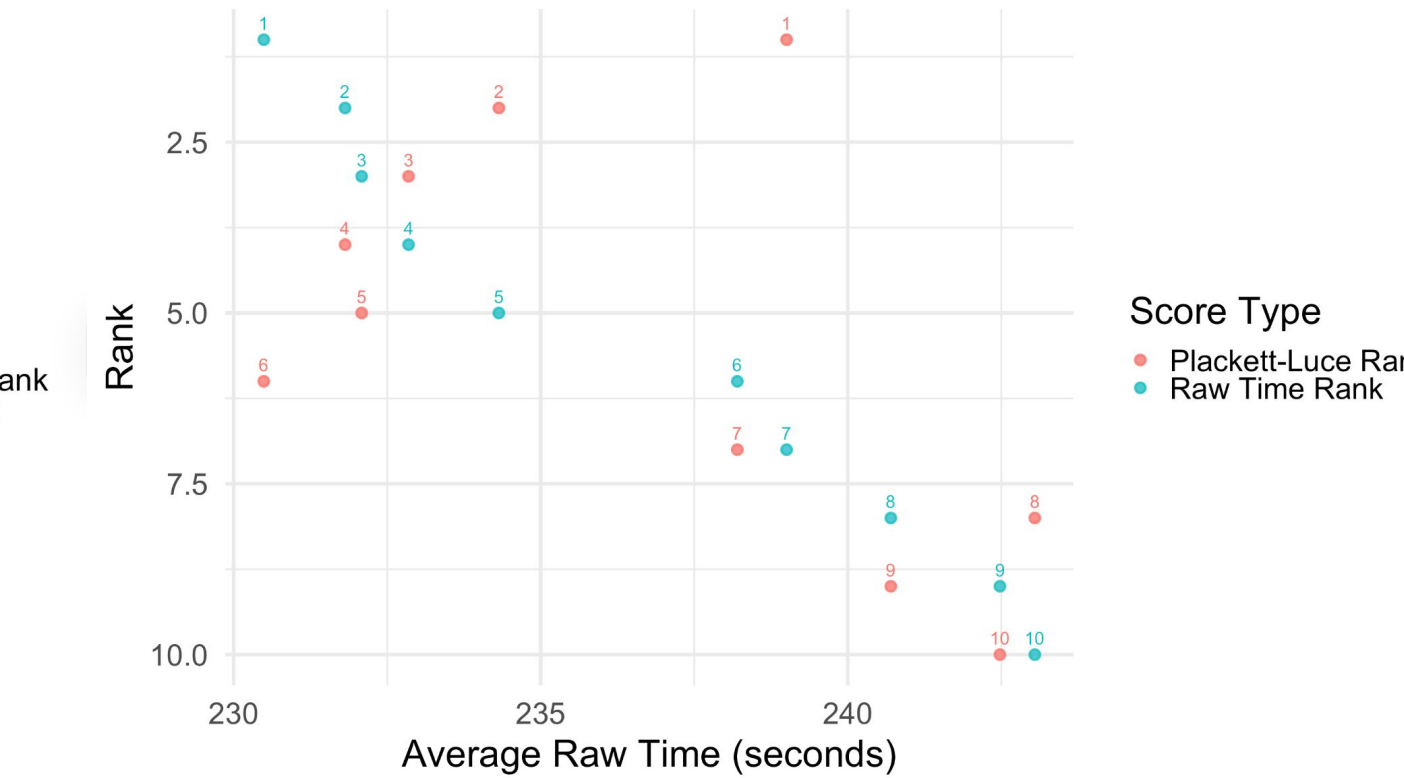
Results

Average Time vs Rank for Top 100 Plackett-Luce Athletes



The figure above displays the top 100 ranked athletes ranked by raw time and Plackett-Luce rankings

Raw Time Rank vs Plackett-Luce Rank (Top 10 Athletes)

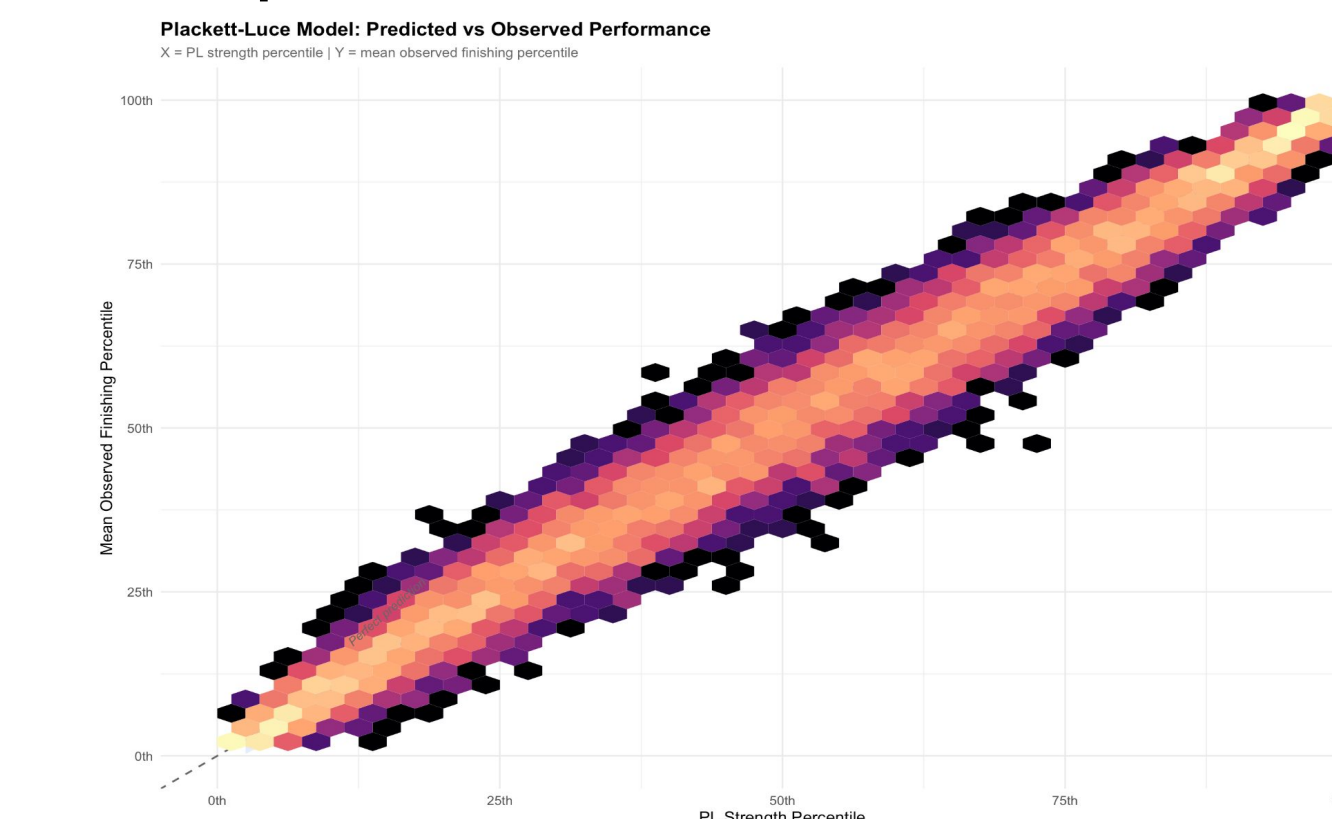


The figure above displays the top 10 ranked athletes ranked by raw time and Plackett-Luce rankings

We compared rankings produced by the Plackett-Luce model with traditional time-based rankings.

There is a clear difference in the way the 2 models have ranked the athletes.

This is expected because time-based rankings treat every performance equally whereas Plackett-Luce takes into account the strength of the competition.



The figure above shows Plackett-Luce estimated strength percentile versus mean observed finishing percentile

The tight cluster of athletes along the diagonal indicates the model successfully ranks the athletes according to their observed finishing percentile across races.

Showing the Plackett-Luce is successful at creating a network of competitive relationships

Conclusions & Future Directions

key takeaways

- Because athletes enter different subsets of races you can't just average finish places.
- An athlete gains more credit for beating stronger competitors.

Probabilistic ranking models provide a principled and scalable framework for evaluating athletic performance in large and fragmented datasets. By incorporating competitive relationships between athletes, this approach enables more meaningful and context-aware comparisons than traditional time-based methods.

Limitations

- Uneven number of races per athlete
- Variation in race conditions and distances
- Incomplete connectivity across all competitors
- Data quality depends on available results

Acknowledgements

Stephanie. (2024, August 11). *Plackett-Luce Model - Statistics How To*. Statistics How To.

<https://www.statisticshowto.com/plackett-luce-model/>

